

Shapiro's Inequality: A Remarkable Use of Convex Functions

Daniel Potievsky



1 Introduction

This article discusses an interesting inequality conjecture that was originally published by the American mathematician Harold Shapiro in *The American Mathematical Monthly* in 1954. It attracted the attention of many renowned mathematicians around the world and was eventually resolved by Vladimir Drinfeld, the 1990 Fields Medal winner, at 17 years old. The statement of the inequality is as follows:

$$\frac{x_1}{x_2 + x_3} + \frac{x_2}{x_3 + x_4} + \dots + \frac{x_{n-1}}{x_n + x_1} + \frac{x_n}{x_1 + x_2} \geq \frac{n}{2}, \quad x_i > 0.$$

2 The Early Cases

The story of this inequality begins much earlier, with the publication of Nesbitt's Inequality in 1902. Nesbitt's Inequality states that for positive numbers a , b , and c ,

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

This is simply the case of $n = 3$ for the Shapiro Inequality. Over the years, many proofs have been developed for this inequality, and perhaps one of the most insightful is through Jensen's Inequality for convex functions. This inequality states that if a function Φ is convex in some neighborhood I ,

then

$$\lambda_1 \Phi(x_1) + \lambda_2 \Phi(x_2) + \dots + \lambda_n \Phi(x_n) \geq \Phi(\lambda_1 x_1 + \lambda_2 x_2 + \dots + \lambda_n x_n)$$

when $\lambda_1 + \lambda_2 + \dots + \lambda_n = 1$ and $x_1, x_2, \dots, x_n \in I$.

2.1 Nesbitt's Inequality

First, consider the proof of Nesbitt's Inequality:

Let $a + b + c = 1$. Fixing the sum in this manner is permissible, because this inequality is homogeneous, meaning that if we multiply each of a , b , and c by some constant ζ , then

$$\frac{\zeta a}{\zeta b + \zeta c} + \frac{\zeta b}{\zeta c + \zeta a} + \frac{\zeta c}{\zeta a + \zeta b} = \frac{a}{b + c} + \frac{b}{c + a} + \frac{c}{a + b},$$

and the inequality does not change, since ζ can be divided out. Therefore, if we let the sum $a + b + c = S$ for some arbitrary S , then we can always let $\zeta = S$, which does not change the inequality but permits $Sa + Sb + Sc = S$ or $a + b + c = 1$.

Therefore, we obtain that $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = \frac{a}{1-a} + \frac{b}{1-b} + \frac{c}{1-c}$.

Let $f(x) = \frac{1}{1-x}$. Then, $f''(x) = \frac{2}{(1-x)^3}$. By examining where $f''(x) \geq 0$, we can conclude that $f(x)$ is convex on the interval $(0, 1)$. But from the condition of $a + b + c = 1$ and the condition that $a, b, c > 0$, we can assert that $a, b, c \in (0, 1)$. So, we can use Jensen's Inequality. Note that

$$\frac{\frac{a}{1-a} + \frac{b}{1-b} + \frac{c}{1-c}}{3} = \frac{1}{3} \cdot \frac{a}{1-a} + \frac{1}{3} \cdot \frac{b}{1-b} + \frac{1}{3} \cdot \frac{c}{1-c} \geq \frac{\frac{a}{3} + \frac{b}{3} + \frac{c}{3}}{1 - (\frac{a}{3} + \frac{b}{3} + \frac{c}{3})} = \frac{1}{2}$$

by Jensen's Inequality. Therefore,

$$\frac{\frac{a}{1-a} + \frac{b}{1-b} + \frac{c}{1-c}}{3} \geq \frac{1}{2},$$

which means

$$\frac{a}{1-a} + \frac{b}{1-b} + \frac{c}{1-c} = \frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2},$$

completing the proof.

2.2 Mordell's Inequality

But the fight for a generalization of this inequality continued. In 1958, the Cambridge mathematician Louis Mordell managed to prove the corresponding inequalities for 4, 5, and 6 variables using a more refined technique with Jensen's Inequality. This technique can be illustrated with the case of 4 variables:

2.2.1 Claim

$$\frac{a}{b+c} + \frac{b}{c+d} + \frac{c}{d+a} + \frac{d}{a+b} \geq 2 \text{ for positive reals } a, b, c, \text{ and } d.$$

2.2.2 Proof

First, let us write out Jensen's Inequality for $f(x) = \frac{1}{x}$ (since $f''(x) = \frac{2}{x^3}$, this function is convex over the set of all positive reals). We obtain:

$$\lambda_1 x_1^{-1} + \lambda_2 x_2^{-1} + \dots + \lambda_n x_n^{-1} \geq \frac{1}{\lambda_1 x_1 + \lambda_2 x_2 + \dots + \lambda_n x_n}.$$

But since $\lambda_1 + \lambda_2 + \dots + \lambda_n = 1$ by the condition for Jensen's Inequality, we can rewrite this as

$$\frac{\lambda_1 x_1^{-1} + \lambda_2 x_2^{-1} + \dots + \lambda_n x_n^{-1}}{\lambda_1 + \lambda_2 + \dots + \lambda_n} \geq \frac{\lambda_1 + \lambda_2 + \dots + \lambda_n}{\lambda_1 x_1 + \lambda_2 x_2 + \dots + \lambda_n x_n}$$

or

$$\frac{\sum \lambda_i x_i^{-1}}{\sum \lambda_i} \geq \frac{\sum \lambda_i}{\sum \lambda_i x_i}.$$

From here, we obtain that

$$\sum \lambda_i x_i^{-1} \geq \frac{(\sum \lambda_i)^2}{\sum \lambda_i x_i}.$$

Returning to the problem with four variables, make the following substitutions: $\lambda_1 = a$, $\lambda_2 = b$, $\lambda_3 = c$, $\lambda_4 = d$. Once again, this is possible because the inequality is homogeneous and we can always say that $a + b + c + d = 1$. We can also let $x_1 = b + c$, $x_2 = c + d$, $x_3 = d + a$, $x_4 = a + b$. Using the inequality that was derived from Jensen's Inequality for four variables, we get

$$\frac{a}{b+c} + \frac{b}{c+d} + \frac{c}{d+a} + \frac{d}{a+b} \geq \frac{(a+b+c+d)^2}{a(b+c) + b(c+d) + c(d+a) + d(a+b)}.$$

Therefore, it is sufficient to prove that

$$\frac{(a+b+c+d)^2}{a(b+c) + b(c+d) + c(d+a) + d(a+b)} \geq 2.$$

Opening parentheses and clearing denominators, we obtain:

$$\begin{aligned} (a+b+c+d)^2 &\geq 2(ab+2ac+ad+bc+2bd+cd) \\ (a^2+b^2+c^2+d^2) + 2(ab+ac+ad+bc+bd+cd) &\geq 2(ab+2ac+ad+bc+2bd+cd) \\ (a^2+b^2+c^2+d^2) &\geq 2ac+2bd \\ (a-c)^2 + (b-d)^2 &\geq 0, \end{aligned}$$

which is clearly true.

2.3 Exercise

Prove the analogous statements for five and six variables.

Hint: The final step will require the Cauchy-Schwarz Inequality.

3 Theorems

Unfortunately, the problem began for $n = 7$, which did not yield to Mordell's technique. Over the years, many attempts were made to resolve the Shapiro Inequality, and there was a certain degree of success achieved with individual cases. In 1961, the case $n = 8$ was resolved; in 1967, the case $n = 10$ was resolved; and in 1974, the case $n = 12$ was resolved. But the questions of whether or not the inequality was true for all n and whether the conjecture was right at all remained.

It turned out that the cyclic nature of this inequality was a great help for conducting further proofs. Denote for simplicity

$$f_n(x_1, x_2, \dots, x_n) = \frac{x_1}{x_2 + x_3} + \frac{x_2}{x_3 + x_4} + \dots + \frac{x_{n-1}}{x_n + x_1} + \frac{x_n}{x_1 + x_2}.$$

We can see by inspecting that this expression is cyclic, because

$$f_n(x_1, x_2, \dots, x_n) = f_n(x_k, x_{k+1}, \dots, x_n, x_1, x_2, \dots, x_{k-1}).$$

3.1 Theorem 1

3.1.1 Claim

If Shapiro's Inequality is untrue for m variables, then it is untrue for $m + 2$ variables.

3.1.2 Proof

Suppose for some m , $f_m(x_1, x_2, \dots, x_m) < \frac{m}{2}$. One can then construct a set of $m + 2$ variables such that $f_m(x_1, x_2, \dots, x_{m+2}) < \frac{m+2}{2}$ as well, which completes the proof. For the proof, use

$$\begin{aligned} f_{m+2}(x_1, x_2, x_3, \dots, x_m, x_1, x_2) &= \frac{x_1}{x_2 + x_3} + \frac{x_2}{x_3 + x_4} + \dots + \frac{x_m}{x_1 + x_2} + \frac{x_1}{x_2 + x_1} + \frac{x_2}{x_1 + x_2} \\ &= f_m(x_1, x_2, \dots, x_m) + \frac{x_1}{x_1 + x_2} + \frac{x_2}{x_1 + x_2} \\ &= f_m(x_1, x_2, \dots, x_m) + 1. \end{aligned}$$

Therefore,

$$f_{m+2}(x_1, x_2, x_3, \dots, x_m, x_1, x_2) = f_m(x_1, x_2, \dots, x_m) + 1 < \frac{m}{2} + 1 = \frac{m+2}{2},$$

or

$$f_{m+2}(x_1, x_2, x_3, \dots, x_m, x_1, x_2) < \frac{m+2}{2},$$

which concludes the proof.

3.2 Theorem 2

3.2.1 Claim

If Shapiro's Inequality is untrue for m variables, where m is an odd integer, then it is untrue for all integers greater than m as well.

3.2.2 Proof

First note the following identity:

$$f_{m+1}(x_1, x_2, \dots, x_i, x_i, \dots, x_m, x_{m+1}) - f_m(x_1, x_2, \dots, x_m) - \frac{1}{2} = \frac{(x_i - x_{i-1})(x_i - x_{i+1})}{2x_i(x_i + x_{i+1})}.$$

This is true because the left side can be rewritten as

$$\begin{aligned} & \left(\frac{x_1}{x_2 + x_3} + \dots + \frac{x_{i-1}}{x_i + x_i} + \frac{x_i}{x_i + x_{i+1}} + \frac{x_i}{x_{i+1} + x_{i+2}} + \dots + \frac{x_m}{x_1 + x_2} \right) - \left(\frac{x_1}{x_2 + x_3} + \dots + \frac{x_m}{x_1 + x_2} \right) - \frac{1}{2} \\ &= \frac{x_{i-1}}{x_i + x_i} + \frac{x_i}{x_i + x_{i+1}} - \frac{x_{i-1}}{x_i + x_{i+1}} - \frac{1}{2}, \end{aligned}$$

with the rest of the terms cancelling. Bringing to a common denominator, this expression equals

$$\begin{aligned} \frac{x_{i-1}(x_i + x_{i+1}) + 2x_i^2 - 2x_{i-1}x_i - x_i(x_i + x_{i+1})}{2x_i(x_i + x_{i+1})} &= \frac{x_{i-1}x_{i+1} + x_i^2 - x_{i-1}x_i - x_ix_{i+1}}{2x_i(x_i + x_{i+1})} \\ &= \frac{(x_i - x_{i-1})(x_i - x_{i+1})}{2x_i(x_i + x_{i+1})}, \end{aligned}$$

as needed.

But in the case that m is odd, note that it is impossible for $(x_i - x_{i-1})(x_i - x_{i+1}) > 0$ for all i . This can be shown by contradiction. If the previous claim were true, then

$$\begin{cases} (x_2 - x_1)(x_2 - x_3) > 0, \\ (x_3 - x_2)(x_3 - x_4) > 0, \\ \vdots \\ (x_m - x_{m-1})(x_m - x_1) > 0, \\ (x_1 - x_m)(x_1 - x_2) > 0. \end{cases}$$

By pairing up the terms that are negatives of each other, multiplying these inequalities out yields that $-(x_1 + x_2)^2(-(x_2 + x_3)^2)\dots(-(x_m + x_1)^2) > 0$. But since there are m parentheses and m is odd, this yields that a negative number is greater than 0, which is a contradiction. Therefore, if m is odd, then there must exist such an i that $(x_i - x_{i-1})(x_i - x_{i+1}) \leq 0$. But using the identity derived earlier, that would mean that

$$f_{m+1}(x_1, x_2, \dots, x_i, x_i, \dots, x_m, x_{m+1}) - f_m(x_1, x_2, \dots, x_m) - \frac{1}{2} \leq 0,$$

or

$$f_{m+1}(x_1, x_2, \dots, x_i, x_i, \dots, x_m, x_{m+1}) \leq f_m(x_1, x_2, \dots, x_m) + \frac{1}{2} < \frac{m+1}{2}$$

by the problem statement. But this actually concludes the proof. If the inequality is untrue for some odd m , then it is also untrue for the even number $m+1$. Using Theorem 1, we can conclude, with an induction over the evens and odds, that the inequality is untrue for all integers greater than m .

4 Resolution

The two previous theorems, though simple, were pivotal to the resolution of the problem. In 1971, for instance, a counterexample was found for the case of $n = 25$. By Theorems 1 and 2, this implies that the inequality is wrong for all $n \geq 25$. In 1985, the mathematician Troesch found a counterexample for $n = 14$, and in 1989, he published a paper that showed that the inequality was true for all odd values up to 23. Thus, in some sense, with much manual work, Shapiro's Inequality was resolved to work only for even values less than or equal to 12 and odd values less than or equal to 23. But the difficulty that mathematicians had with finding counterexamples was very curious, and Shapiro's conjecture of $\frac{n}{2}$ clearly worked for many of the smaller cases. Perhaps the problem needed to be modified.

5 The Question of a Coefficient

In particular, the question arose as to whether the Shapiro Inequality was always true for some coefficient γ , namely:

$$\frac{x_1}{x_2 + x_3} + \frac{x_2}{x_3 + x_4} + \dots + \frac{x_{n-1}}{x_n + x_1} + \frac{x_n}{x_1 + x_2} \geq \gamma \cdot \frac{n}{2}$$

The process of experimenting and seeking out the value of γ lasted many years.

In fact, the following problem appeared in the Third Soviet Mathematics Olympiad in 1969:

5.0.1 Claim

Prove that if all $x_i > 0$, then $\frac{x_1}{x_2 + x_3} + \frac{x_2}{x_3 + x_4} + \dots + \frac{x_{n-1}}{x_n + x_1} + \frac{x_n}{x_1 + x_2} \geq \frac{n}{4}$.

5.0.2 Proof

This inequality is cyclic, so we can assume without loss of generality that x_1 is the maximum of all the x_i . Denote that maximum to be a_1 . Then, implement the following algorithm: look in the denominator of that fraction and find the larger of the two numbers there (x_2 and x_3). Select that maximum number, and denote it a_2 . Then, find the fraction that has this maximum in its numerator. (This will be $\frac{x_2}{x_3 + x_4}$ or $\frac{x_3}{x_4 + x_5}$, depending on whether x_2 or x_3 is larger.) Look in the denominator of that fraction, and denote the maximum of the two numbers there as a_3 , and so on.

Continue this process until step k : reaching an a_k such that the maximum of the next two numbers is $x_1 = a_1$. How is it known that the algorithm will reach x_1 eventually? Looking at the algorithm in action will show that it moves either by 1 or 2 fractions at a time, since after selecting a fraction with x_i in the numerator, the next fraction that will be selected will either have x_{i+1} or x_{i+2} in the numerator. Since x_1 is the maximum of all the x_i and it is contained in the denominator of two consecutive fractions, $\frac{x_{n-1}}{x_n + x_1}$ and $\frac{x_n}{x_1 + x_2}$, the algorithm will not miss it, since it moves by at most 2 fractions. And because the algorithm moves by at most 2 fractions, it can be asserted that $k \geq \frac{n}{2}$, since the minimal number of steps that will complete this algorithm occurs when the step of the algorithm jumps by 2 fractions, not 1, every time.

As such, it can be stated that

$$\frac{x_1}{x_2 + x_3} + \frac{x_2}{x_3 + x_4} + \dots + \frac{x_{n-1}}{x_n + x_1} + \frac{x_n}{x_1 + x_2} \geq \frac{a_1}{2a_2} + \frac{a_2}{2a_3} + \dots + \frac{a_k}{2a_1}.$$

From here, we use the Inequality of Arithmetic and Geometric Means (AM-GM) to get

$$\frac{a_1}{2a_2} + \frac{a_2}{2a_3} + \dots + \frac{a_k}{2a_1} \geq k \cdot \sqrt[k]{\frac{a_1}{2a_2} \cdot \frac{a_2}{2a_3} \cdot \dots \cdot \frac{a_k}{2a_1}} = \frac{k}{2} \geq \frac{n}{4}.$$

The proof is complete. (This is essentially a proof for $\gamma = \frac{1}{2}$.)

5.1 Exercises

1. Prove that $f_n(x_1, x_2, \dots, x_n) \geq (\sqrt{2} - 1)n \approx 0.41n$.

2. Prove that $f_n(x_1, x_2, \dots, x_n) \geq \frac{5}{12}n$.

Hint: Use the identity $\frac{x_i}{x_{i+1}+x_{i+2}} + \alpha = \frac{x_i+\beta x_{i+1}}{x_{i+1}+x_{i+2}} + \frac{\alpha(\beta x_{i+1}+x_{i+2})}{x_{i+1}+x_{i+2}}$, where $\beta + \alpha\beta = \alpha$, or $\beta = \frac{\alpha}{\alpha+1}$. Proceed with a manipulation using AM-GM. The exercises require 2 different values of α .

6 Drinfeld's Final Resolution to the Shapiro Inequality

Suppose $\frac{x_1}{x_2} = k_1, \frac{x_2}{x_3} = k_2, \dots, \frac{x_n}{x_1} = k_n$. Note that $k_1 k_2 \dots k_n = 1$. Take all the values k_i and put them into increasing order. Denote the largest as m_1 , the second as m_2 , ..., the smallest as m_n (note: $m_1 m_2 \dots m_n = 1$). Therefore, we can see that the sequences $\{\frac{1}{m_1}, \frac{1}{m_2}, \dots, \frac{1}{m_n}\}$ and $\{\frac{1}{m_1+1}, \frac{1}{m_2+1}, \dots, \frac{1}{m_n+1}\}$ are non-decreasing; that is, $\frac{1}{m_1} \leq \frac{1}{m_2} \leq \dots \leq \frac{1}{m_n}$ and $\frac{1}{m_1+1} \leq \frac{1}{m_2+1} \leq \dots \leq \frac{1}{m_n+1}$. By the rearrangement inequality, it is known that a sum of pairwise products of two ordered sets $a_1 \geq a_2 \geq \dots \geq a_n$ and $b_1 \geq b_2 \geq \dots \geq b_n$ reaches its minimum when it is arranged as $a_1 b_n + a_2 b_{n-1} + \dots + a_{n-1} b_2 + a_n b_1$.

Therefore,

$$\begin{aligned} \frac{x_1}{x_2 + x_3} + \frac{x_2}{x_3 + x_4} + \dots + \frac{x_{n-1}}{x_1 + x_2} + \frac{x_n}{x_1 + x_2} &= \frac{1}{k_1(k_2 + 1)} + \frac{1}{k_2(k_3 + 1)} + \dots + \frac{1}{k_n(k_1 + 1)} \\ &\geq \frac{1}{m_1(m_n + 1)} + \frac{1}{m_2(m_{n-1} + 1)} + \dots + \frac{1}{m_n(m_1 + 1)}, \end{aligned}$$

or

$$f_n \geq \frac{1}{m_1(m_n + 1)} + \frac{1}{m_2(m_{n-1} + 1)} + \dots + \frac{1}{m_n(m_1 + 1)}$$

where the inequality is obtained from the minimality of the rearrangement inequality for the two ordered sets given. Denote then $r_i = \frac{1}{m_i(m_{n-i+1}+1)} + \frac{1}{m_{n-i+1}(m_i+1)}$. With this definition, we obtain an equivalent inequality:

$$f_n \geq \frac{1}{2} \sum_{i=1}^n r_i$$

Let $c_i = m_i \cdot m_{n-i+1}$. Note that $c_1 c_2 \dots c_n = 1$.

Rewriting r_i , we get

$$\begin{aligned}
r_i &= \frac{m_{n-i+1}(m_i + 1) + m_i(m_{n-i+1} + 1)}{m_i m_{n-i+1}(m_i + 1)(m_{n-i+1} + 1)} \\
&= \frac{2c_i + m_i + m_{n-i+1}}{c_i(m_i + 1)(m_{n-i+1} + 1)} \\
&= \frac{1}{c_i} \left(\frac{2c_i + m_i + m_{n-i+1}}{(m_i + 1)(m_{n-i+1} + 1)} \right) \\
&= \frac{1}{c_i} \left(\frac{c_i + m_i \cdot m_{n-i+1} + m_i + m_{n-i+1}}{(m_i + 1)(m_{n-i+1} + 1)} \right) \\
&= \frac{1}{c_i} \left(\frac{c_i - 1 + m_i \cdot m_{n-i+1} + m_i + m_{n-i+1} + 1}{(m_i + 1)(m_{n-i+1} + 1)} \right) \\
&= \frac{1}{c_i} \left(\frac{c_i - 1}{(m_i + 1)(m_{n-i+1} + 1)} + 1 \right).
\end{aligned}$$

By AM-GM,

$$\begin{aligned}
\frac{m_i + m_{n-i+1}}{2} &\geq \sqrt{m_i \cdot m_{n-i+1}} = \sqrt{c_i} \\
m_i + m_{n-i+1} &\geq 2\sqrt{c_i}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
m_i \cdot m_{n-i+1} + m_i + m_{n-i+1} + 1 &\geq c_i + 2\sqrt{c_i} + 1 \\
(m_i + 1)(m_{n-i+1} + 1) &\geq (1 + \sqrt{c_i})^2.
\end{aligned}$$

If $c_i < 1$, then $\frac{c_i - 1}{(m_i + 1)(m_{n-i+1} + 1)} < 0$, and this value will attain a maximum when the denominator is maximized. Therefore,

$$r_i = \frac{1}{c_i} \left(\frac{c_i - 1}{(m_i + 1)(m_{n-i+1} + 1)} + 1 \right) \geq \frac{1}{c_i} \left(\frac{c_i - 1}{(1 + \sqrt{c_i})^2} + 1 \right) = \frac{2}{c_i + \sqrt{c_i}}$$

when $c_i < 1$. If $c_i \geq 1$, then

$$r_i = \frac{1}{c_i} \left(\frac{c_i - 1}{(m_i + 1)(m_{n-i+1} + 1)} + 1 \right) \geq \frac{1}{c_i}.$$

From the definition of c_i , it follows that $c_i > 0$, because $m_i, m_{n-i+1} > 0$. Then, the following substitution can be made: $c_i = e^{\eta_i}$. Note that $\eta_1 + \eta_2 + \dots + \eta_n = \ln(c_1 c_2 \dots c_n) = 0$. Therefore, from the previous inequalities, we get that

$$f_n \geq \sum_{c_i \geq 1} e^{-\eta_i} + \sum_{c_i < 1} 2(e^{\frac{\eta_i}{2}} + e^{\eta_i})^{-1}.$$

Let $\psi(x)$ be the largest convex function that does not exceed the functions $\mu(x) = e^{-x}$ and $\theta(x) = 2(e^{\frac{x}{2}} + e^x)^{-1}$.

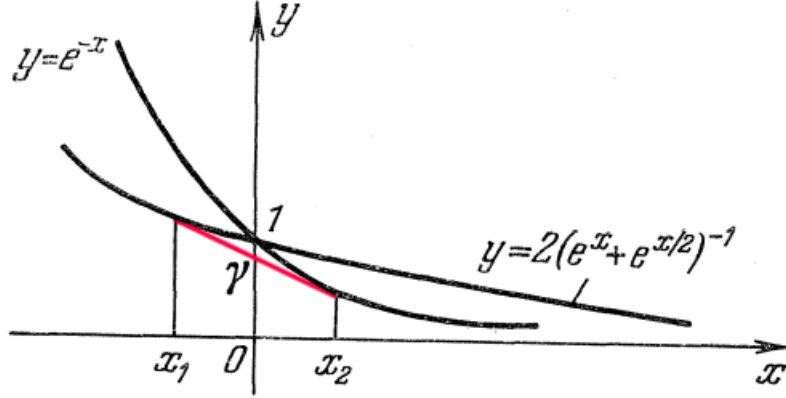


Figure 1: Depiction of $\psi(x)$.

Therefore, assuming that $f_n \geq \gamma \cdot \frac{n}{2}$, we can search for γ by examining $\frac{2}{n}f_n$. Using the last inequality and the definition of $\psi(x)$, we get:

$$\frac{2}{n}f_n \geq \frac{1}{n} \sum_{i=1}^n \psi(\eta_i) \geq \psi\left(\frac{\eta_1 + \eta_2 + \dots + \eta_n}{n}\right) = \psi(0)$$

by Jensen's Inequality. As such, we have illustrated that $\gamma \geq \psi(0)$. But we can find $\psi(0)$ by describing ψ , which was defined as the largest convex function that does not exceed μ and θ . Investigating, we see that μ and θ are convex functions themselves and that $\mu(0) = \theta(0) = 1$. Furthermore, for these functions, $\mu(x) > \theta(x)$ for $x < 0$, and $\mu(x) < \theta(x)$ for $x > 0$. Finding the function ψ means finding the maximum "convex hull," or in this case, a common tangent line $y = mx + b$ for these two functions, connecting $(x_1, 2(e^{\frac{x_1}{2}} + e^{x_1})^{-1})$ and (x_2, e^{-x_2}) , where $x_1 < x_2$. This satisfies the conditions of convexity and maximality of ψ simultaneously, so the function is defined as:

$$\psi(x) = \begin{cases} 2(e^{\frac{x}{2}} + e^x)^{-1} & \text{if } x < x_1 \\ mx + b & \text{if } x_1 \leq x \leq x_2 \\ e^{-x} & \text{if } x > x_2 \end{cases}$$

Finding $mx + b$ ultimately comes down to finding x_1 and x_2 . This can be done by solving the system of equations:

$$\begin{cases} \mu'(x_2) = \theta'(x_1) \\ \mu'(x_2)(x_1 - x_2) + \mu(x_2) = \theta'(x_1) \end{cases}$$

From here, we find that $x_1 = -0.20081\dots$ and $x_2 = 0.155195\dots$. Therefore, $\psi(x) = mx + b$ over $-0.20081\dots \leq x \leq 0.155195\dots$ and $\psi(0) = b$, which can be calculated to be $\gamma = 0.989134\dots$. This is known as Drinfeld's Constant. Therefore, $\frac{2}{n}f_n \geq \gamma \geq \psi(0) \Rightarrow f_n \geq \frac{n}{2} \cdot 0.989134\dots$

(Note: Drinfeld's full method relied on proving the equality $\gamma = \psi(0)$ based on a previous work by Robert Rankin, a Scottish mathematician who examined γ as a sequence for $n \rightarrow \infty$ in a previous paper. He proved that $\lim_{n \rightarrow \infty} \gamma$ converges and that $\lim_{n \rightarrow \infty} \gamma = \inf \gamma$. Since $\inf \gamma$ is really what we are looking for, all that is left is to evaluate $\lim_{n \rightarrow \infty} \gamma$, and Drinfeld proved that $\lim_{n \rightarrow \infty} \gamma = \psi(0)$.)

7 Conclusion

The Shapiro Inequality itself has been resolved through the efforts of many mathematicians over the course of decades. It is worth noting that much of the problem faced by mathematicians originated in the very nature of Drinfeld's Constant, 0.989134.... This number is very close to 1, which explains why mathematicians had such difficulty in finding counterexamples. *Post facto*, we can reason that since Drinfeld's Constant provides a strict lower bound, any contradiction that had been found to Shapiro's Inequality in the past turned out to lie between $\frac{n}{2} \cdot 0.989134\dots$ and $\frac{n}{2}$. And finding the right numbers to fall within that tiny interval is not easy!

References

V. G. Drinfeld, "On a cyclic inequality," Mat. Zametki 9 (1971), 113–119 (in Russian).